

ENM 5310: Data-driven Modeling and Probabilistic Scientific Computing

Lecture #3: Primer on Probability



Probability density functions

$$\Pr[a \leq X \leq b] = \int_a^b f_X(x) dx.$$

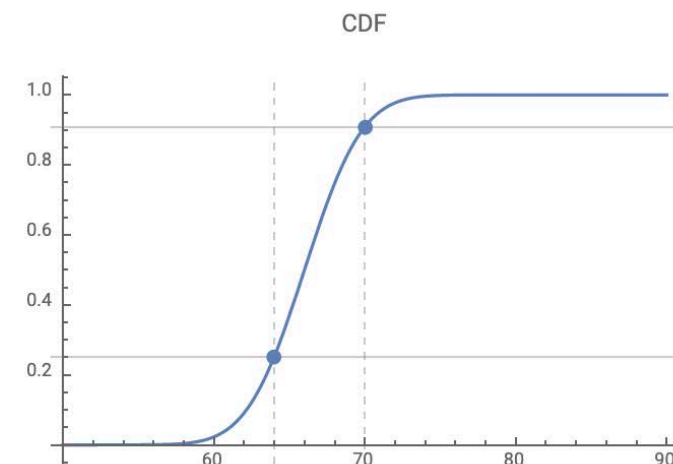
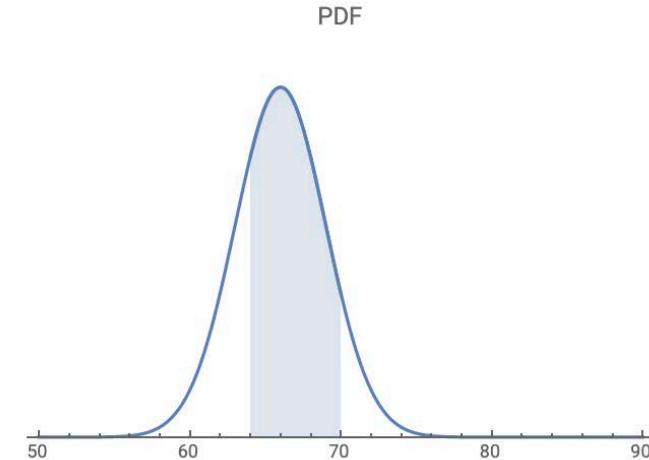
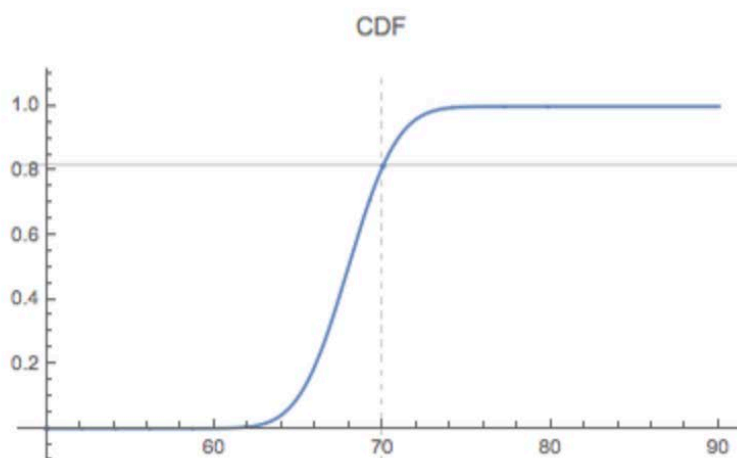
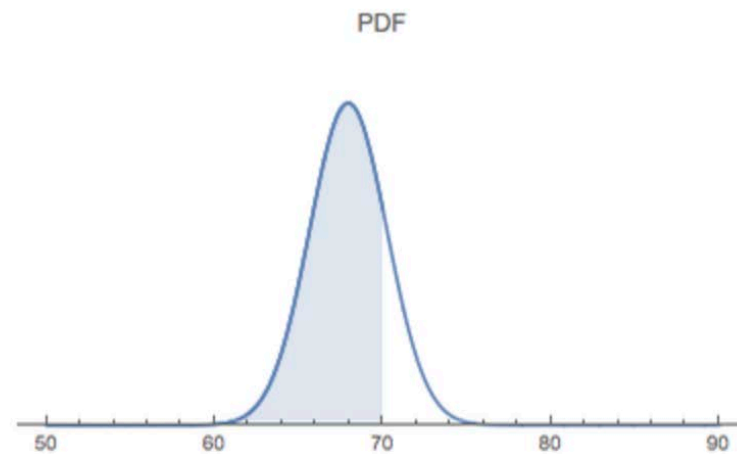
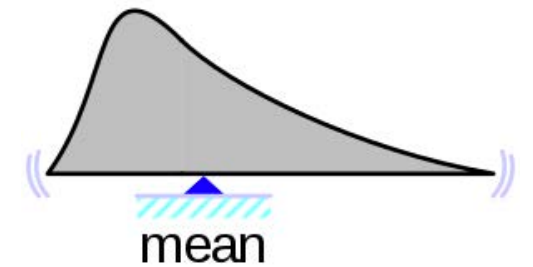
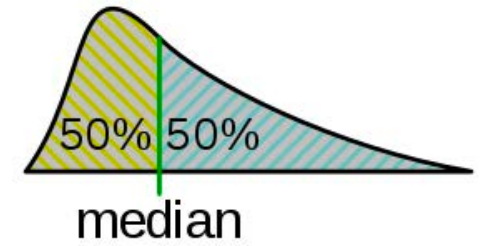
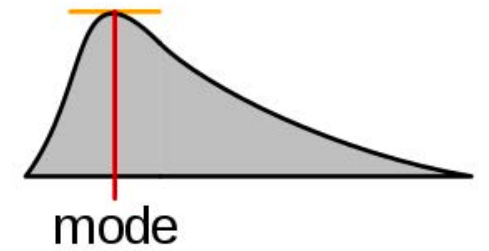
Hence, if F_X is the **cumulative distribution function** of X , then:

$$F_X(x) = \int_{-\infty}^x f_X(u) du,$$

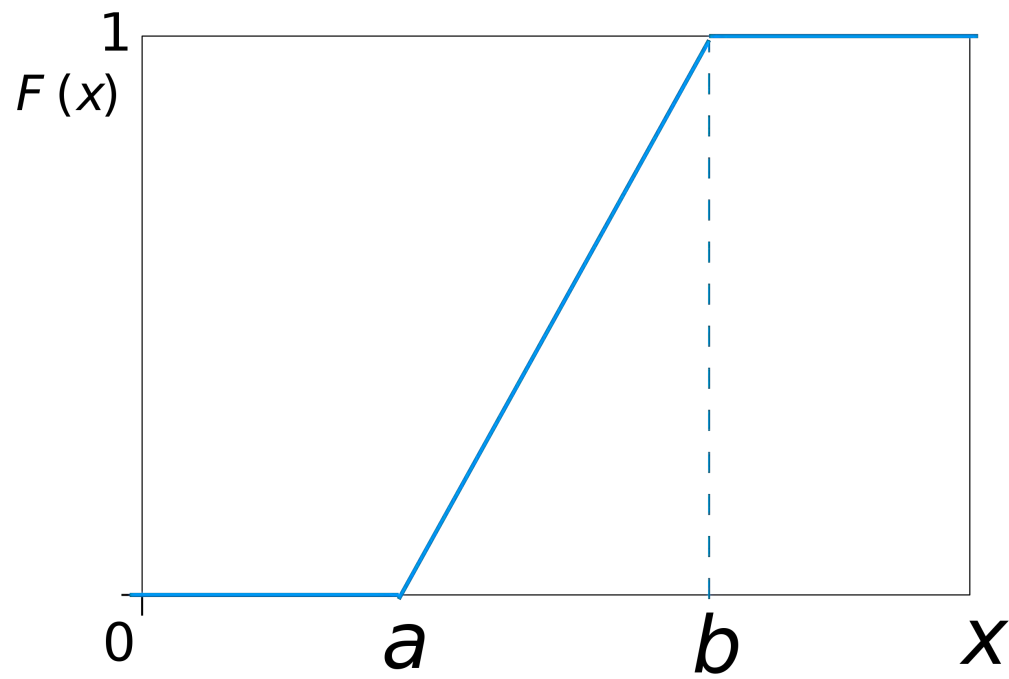
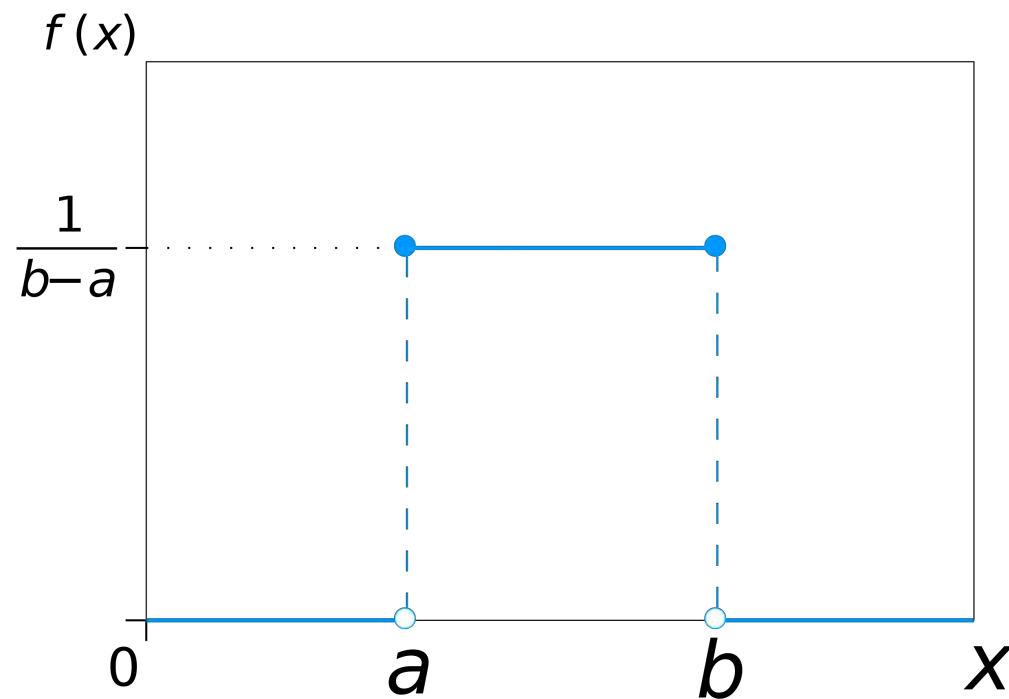
and (if f_X is continuous at x)

$$f_X(x) = \frac{d}{dx} F_X(x).$$

Intuitively, one can think of $f_X(x) dx$ as being the probability of X falling within the infinitesimal **interval** $[x, x + dx]$.

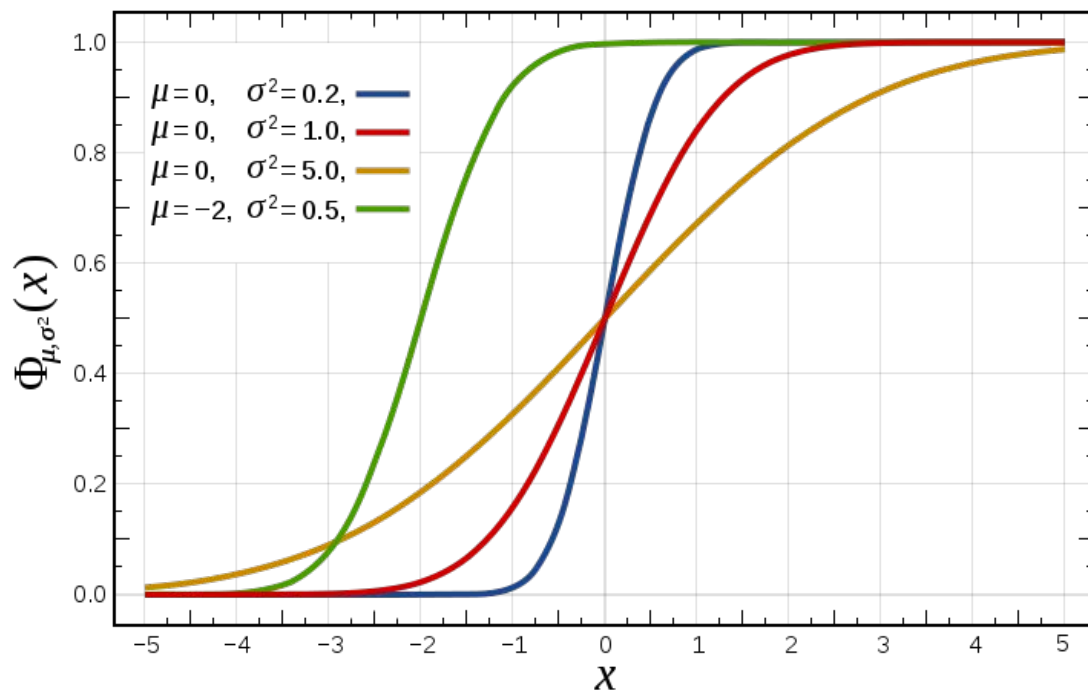
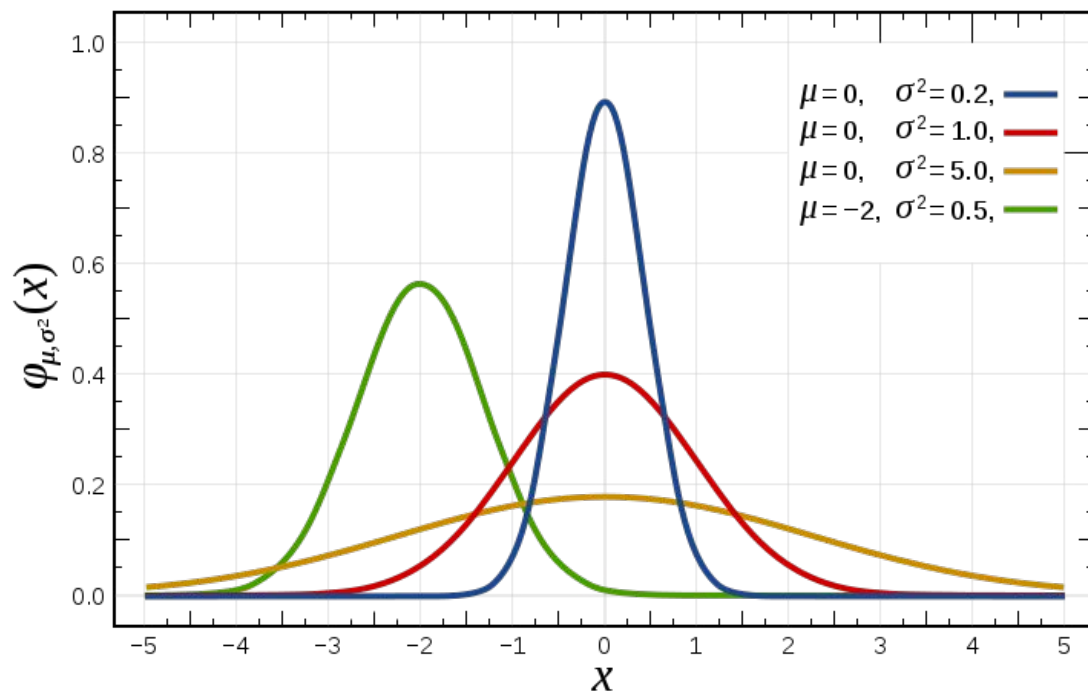


Uniform distribution



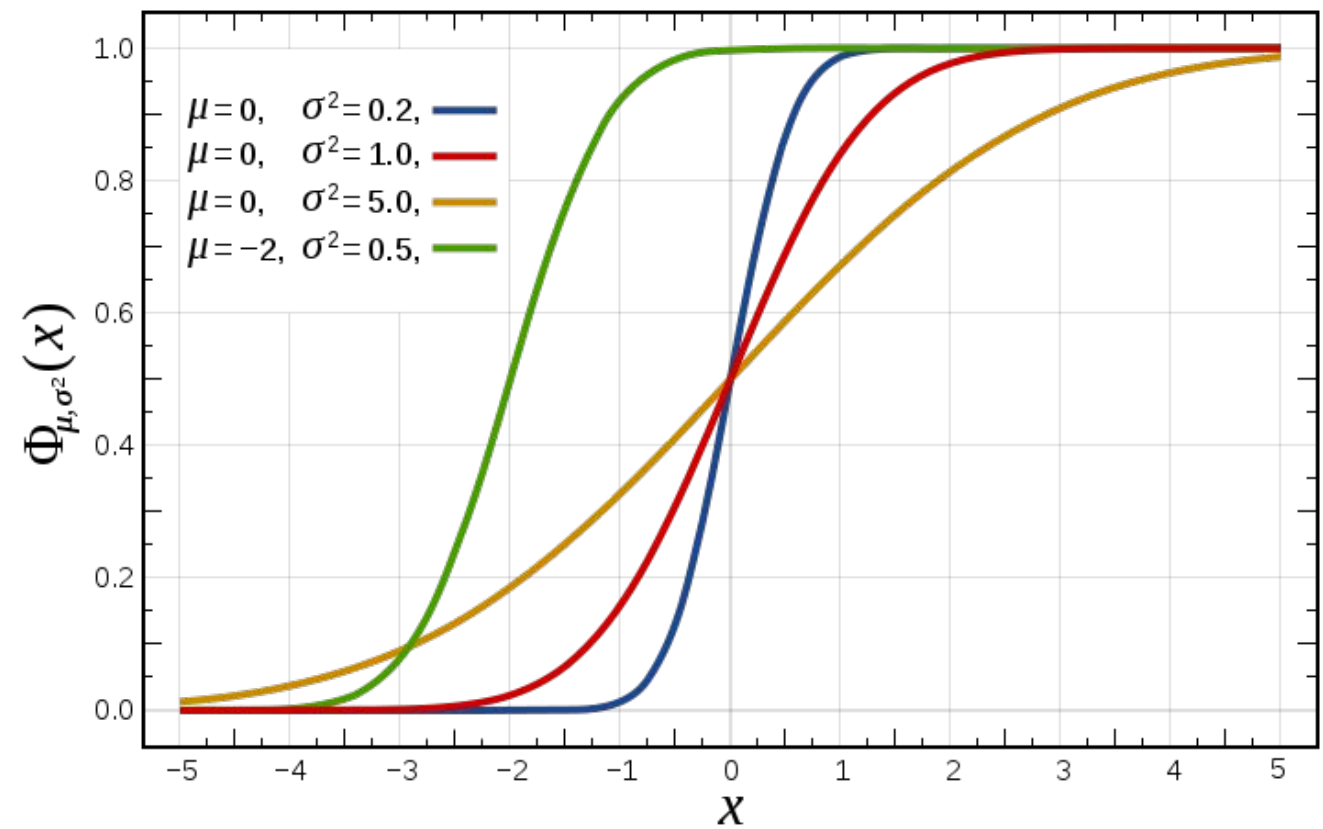
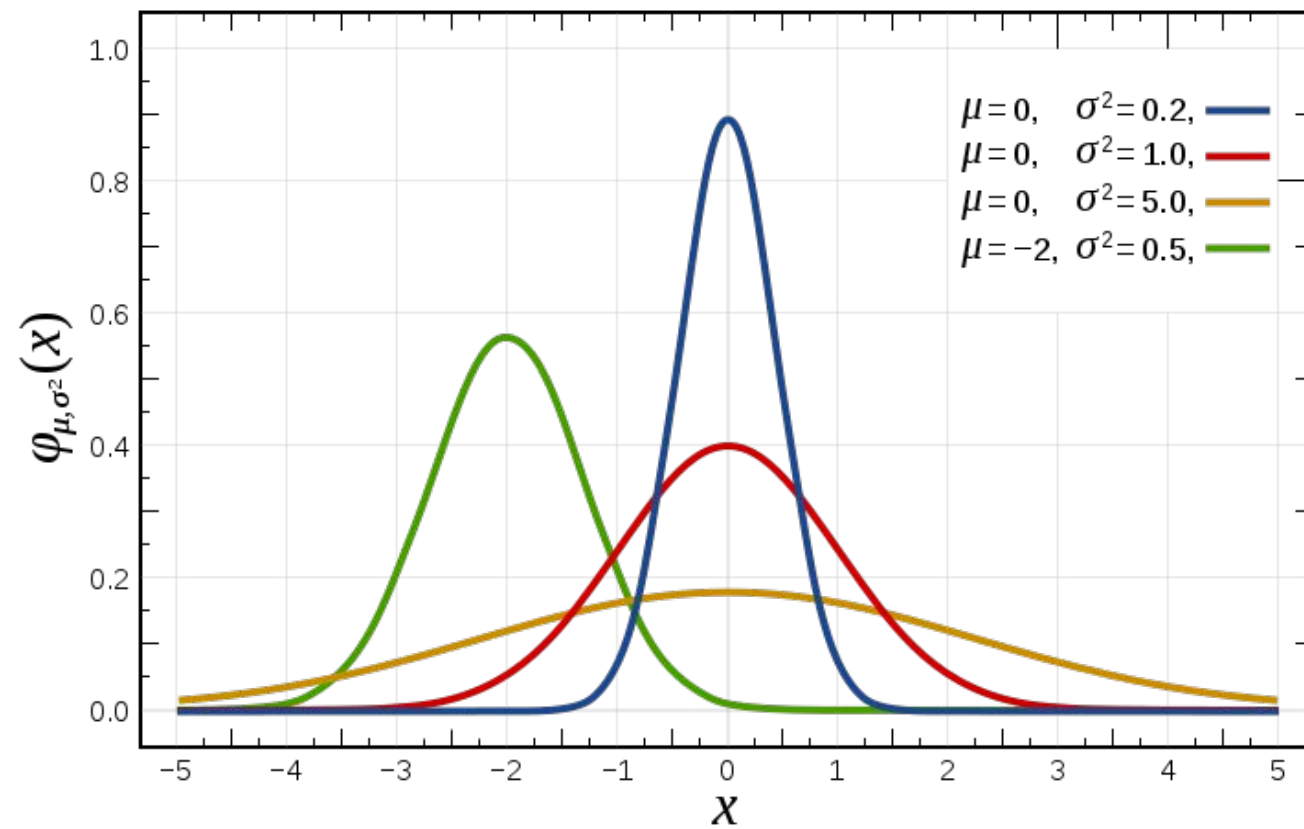
Notation	$\mathcal{U}(a, b)$ or $\text{unif}(a, b)$
Parameters	$-\infty < a < b < \infty$
Support	$x \in [a, b]$
PDF	$\begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$
CDF	$\begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b) \\ 1 & \text{for } x \geq b \end{cases}$
Mean	$\frac{1}{2}(a + b)$
Median	$\frac{1}{2}(a + b)$
Mode	any value in (a, b)
Variance	$\frac{1}{12}(b - a)^2$
Skewness	0
Ex. kurtosis	$-\frac{6}{5}$
Entropy	$\ln(b - a)$
MGF	$\begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)} & \text{for } t \neq 0 \\ 1 & \text{for } t = 0 \end{cases}$
CF	$\begin{cases} \frac{e^{itb} - e^{ita}}{it(b-a)} & \text{for } t \neq 0 \\ 1 & \text{for } t = 0 \end{cases}$

Recap: Univariate Gaussian distribution



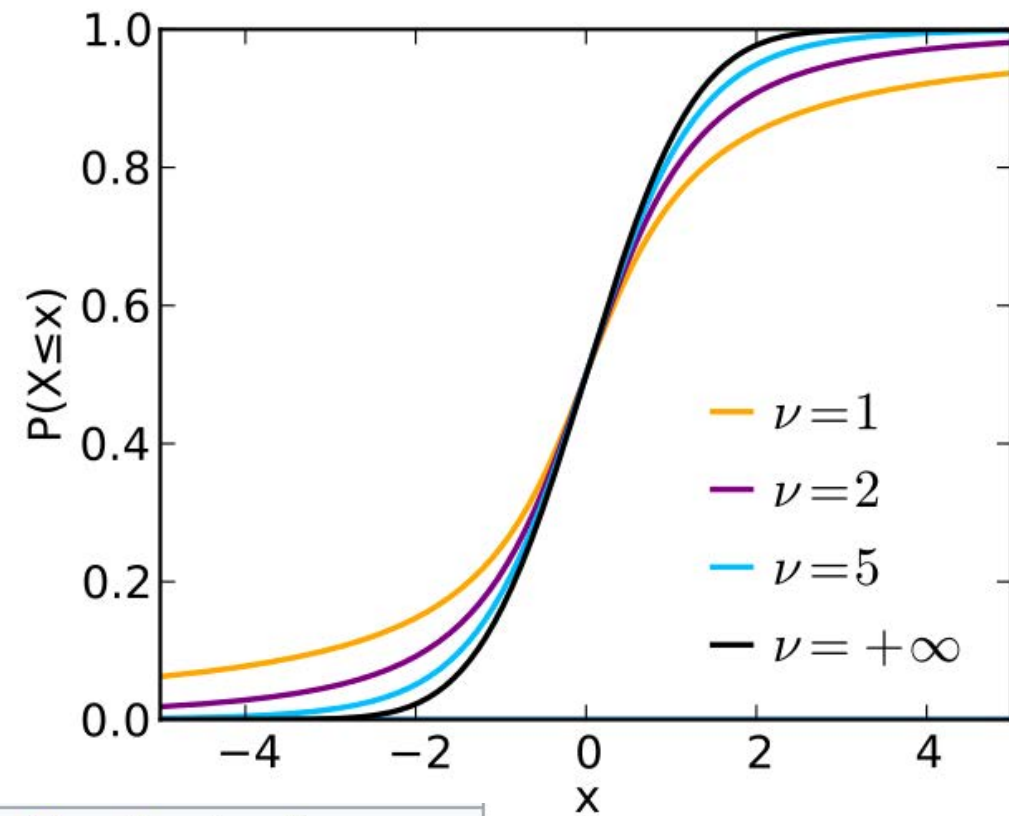
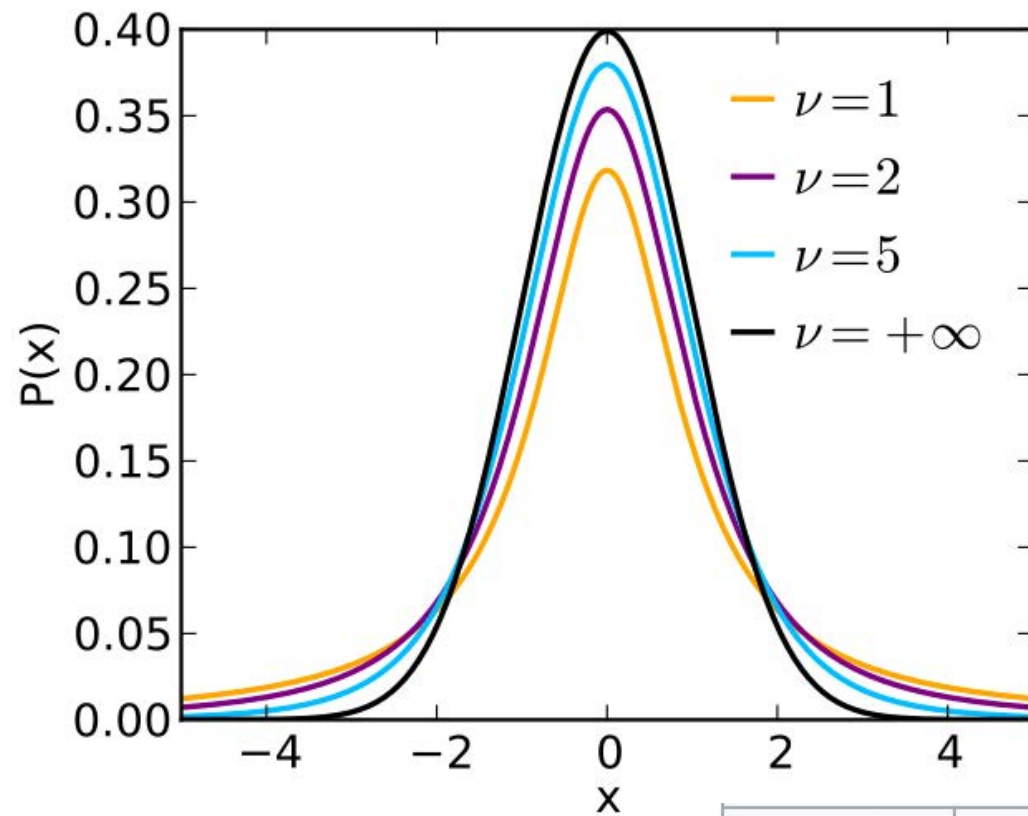
Notation	$\mathcal{N}(\mu, \sigma^2)$
Parameters	$\mu \in \mathbb{R}$ = mean (location) $\sigma^2 > 0$ = variance (squared scale)
Support	$x \in \mathbb{R}$
PDF	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
CDF	$\frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right) \right]$
Quantile	$\mu + \sigma\sqrt{2} \operatorname{erf}^{-1}(2F - 1)$
Mean	μ
Median	μ
Mode	μ
Variance	σ^2
Skewness	0
Ex. kurtosis	0
Entropy	$\frac{1}{2} \log(2\pi e \sigma^2)$
MGF	$\exp(\mu t + \sigma^2 t^2 / 2)$
CF	$\exp(i\mu t - \sigma^2 t^2 / 2)$
Fisher information	$\mathcal{I}(\mu, \sigma) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 2/\sigma^2 \end{pmatrix} \quad \mathcal{I}(\mu, \sigma^2) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}$
Kullback-Leibler divergence	$D_{\text{KL}}(\mathcal{N}_0 \parallel \mathcal{N}_1) = \frac{1}{2} \left\{ (\sigma_0/\sigma_1)^2 + \frac{(\mu_1 - \mu_0)^2}{\sigma_1^2} - 1 + 2 \ln \frac{\sigma_1}{\sigma_0} \right\}$

The Gaussian distribution



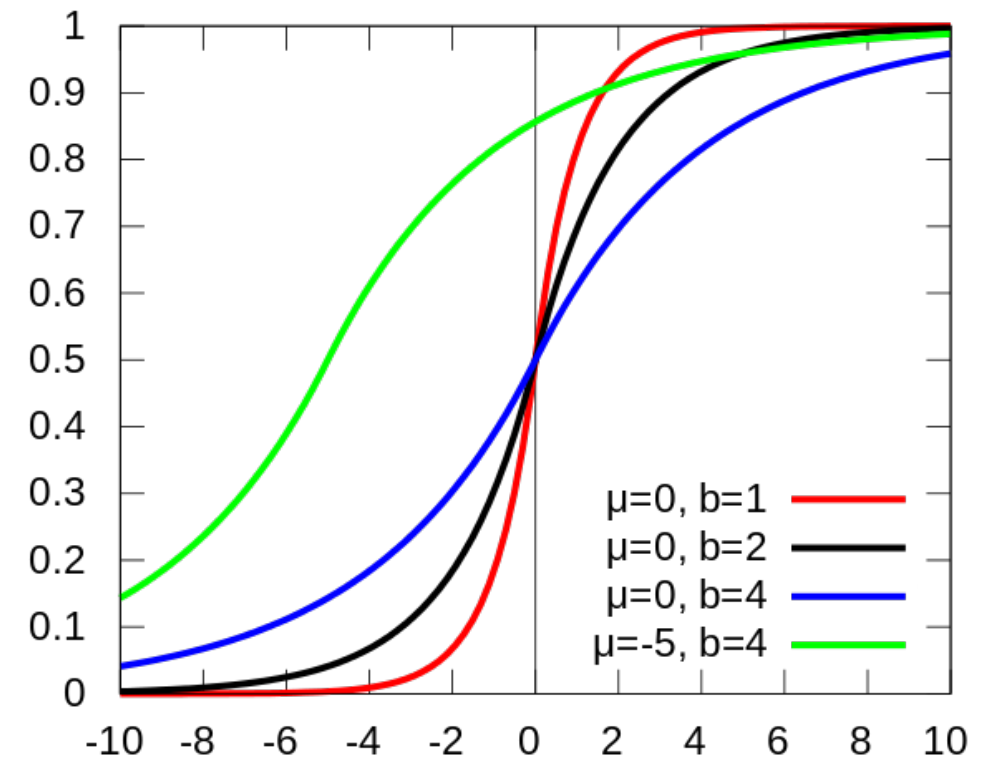
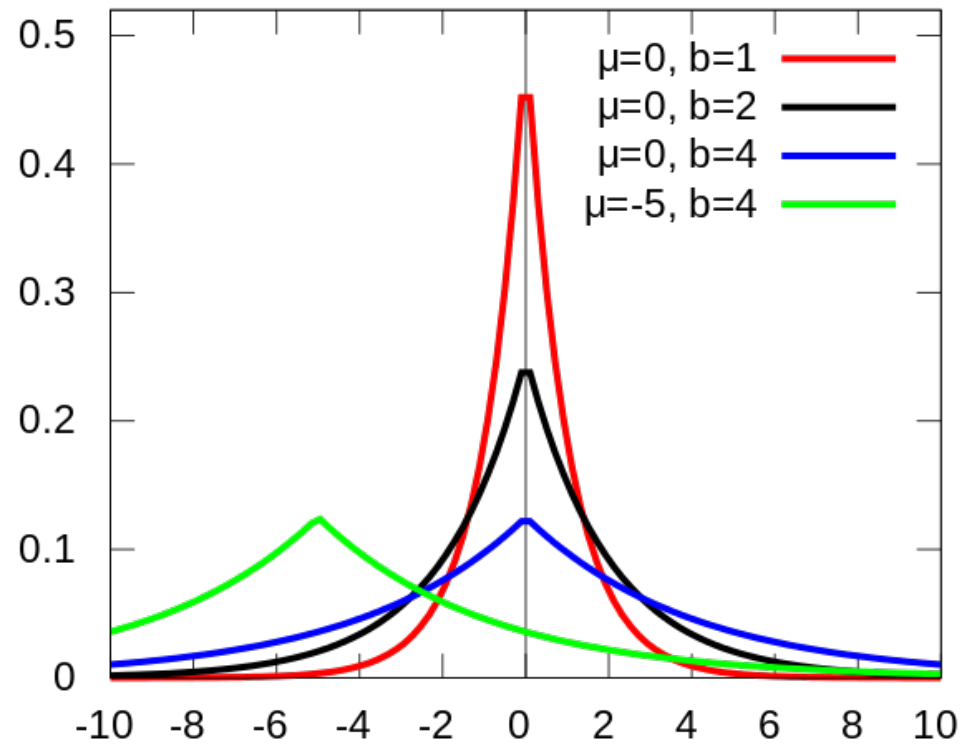
Notation	$\mathcal{N}(\mu, \sigma^2)$
Parameters	$\mu \in \mathbb{R}$ = mean (location) $\sigma^2 > 0$ = variance (squared scale)
Support	$x \in \mathbb{R}$
PDF	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
CDF	$\frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x - \mu}{\sigma\sqrt{2}}\right) \right]$
Quantile	$\mu + \sigma\sqrt{2} \operatorname{erf}^{-1}(2F - 1)$
Mean	μ
Median	μ
Mode	μ
Variance	σ^2

The Student-t distribution



Parameters	$\nu > 0$ degrees of freedom (real)
Support	$x \in (-\infty; +\infty)$
PDF	$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$
CDF	$\frac{1}{2} + x \Gamma\left(\frac{\nu+1}{2}\right) \times$ $\frac{{}_2F_1\left(\frac{1}{2}, \frac{\nu+1}{2}; \frac{3}{2}; -\frac{x^2}{\nu}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)}$ where ${}_2F_1$ is the hypergeometric function
Mean	0 for $\nu > 1$, otherwise undefined
Median	0
Mode	0
Variance	$\frac{\nu}{\nu-2}$ for $\nu > 2$, ∞ for $1 < \nu \leq 2$, otherwise undefined

The Laplace distribution



Parameters	μ location (real) $b > 0$ scale (real)
Support	$x \in (-\infty; +\infty)$
PDF	$\frac{1}{2b} \exp\left(-\frac{ x - \mu }{b}\right)$
CDF	$\begin{cases} \frac{1}{2} \exp\left(\frac{x - \mu}{b}\right) & \text{if } x < \mu \\ 1 - \frac{1}{2} \exp\left(-\frac{x - \mu}{b}\right) & \text{if } x \geq \mu \end{cases}$
Mean	μ
Median	μ
Mode	μ
Variance	$2b^2$

Gaussian vs Student-t vs Laplace

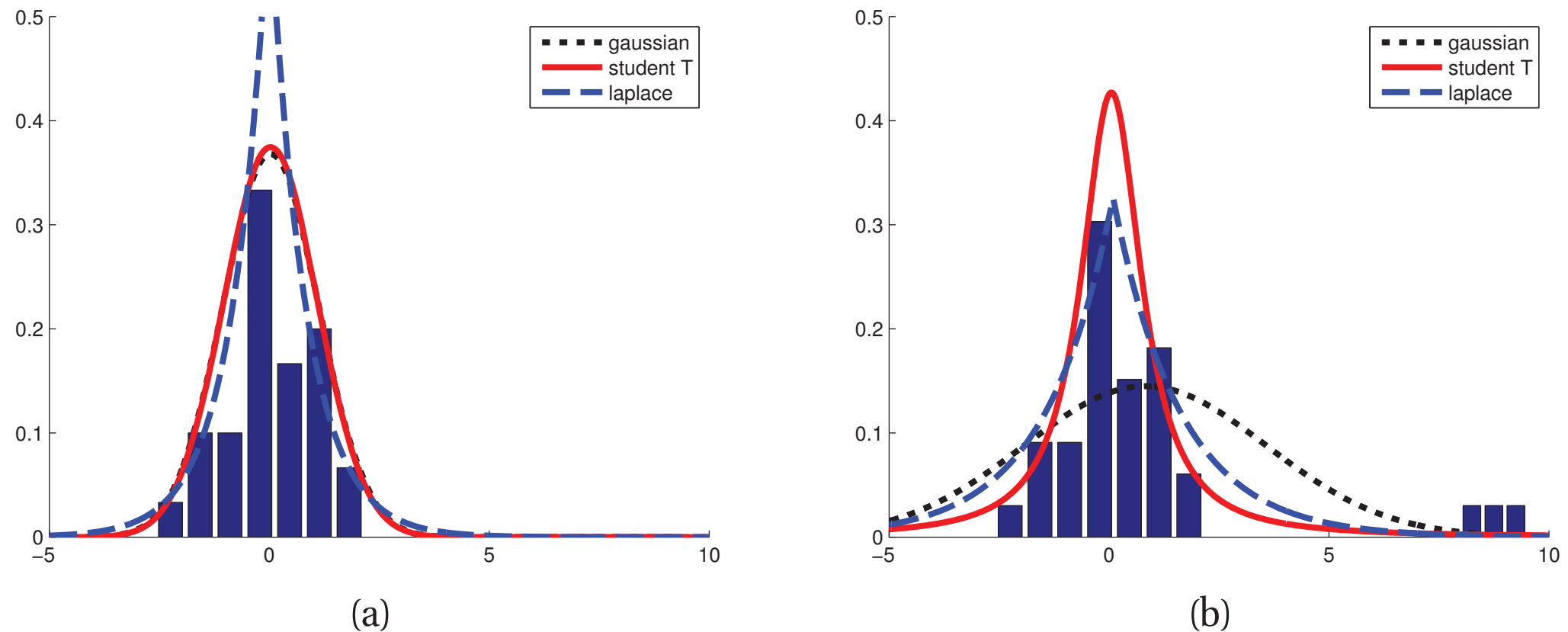
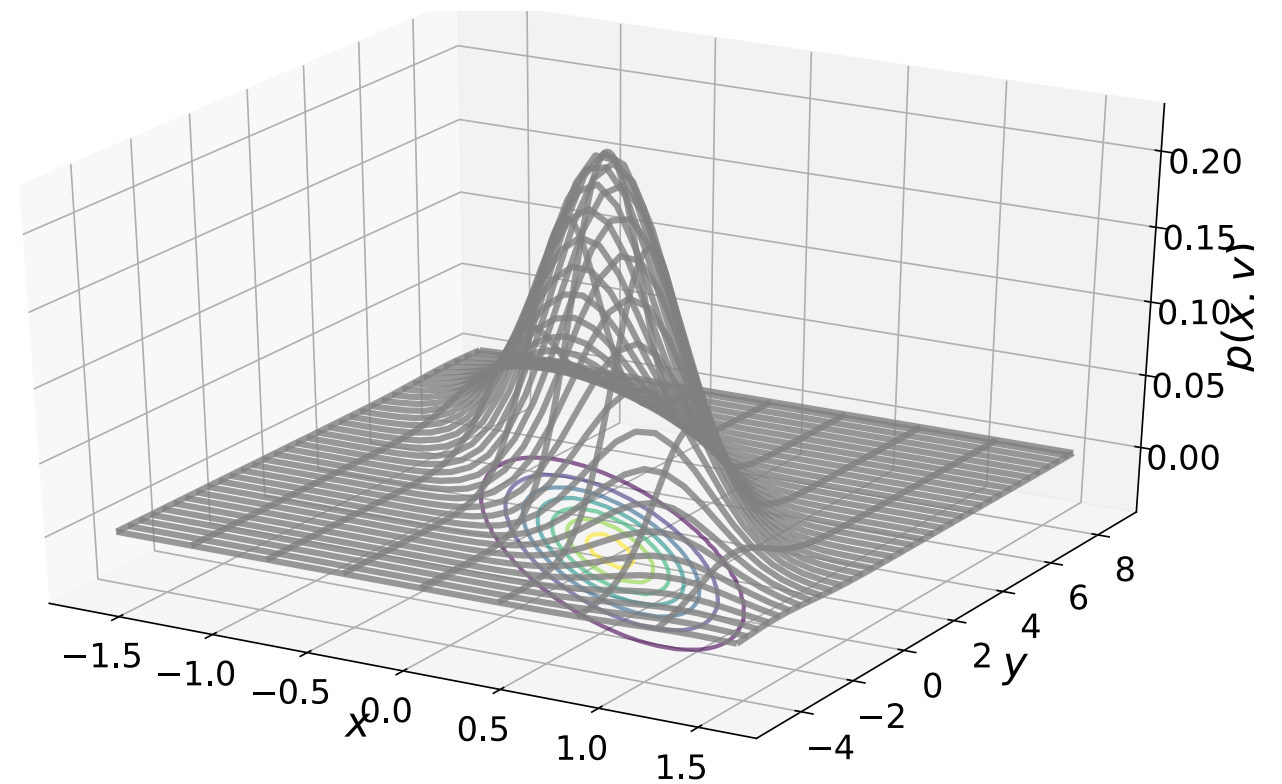
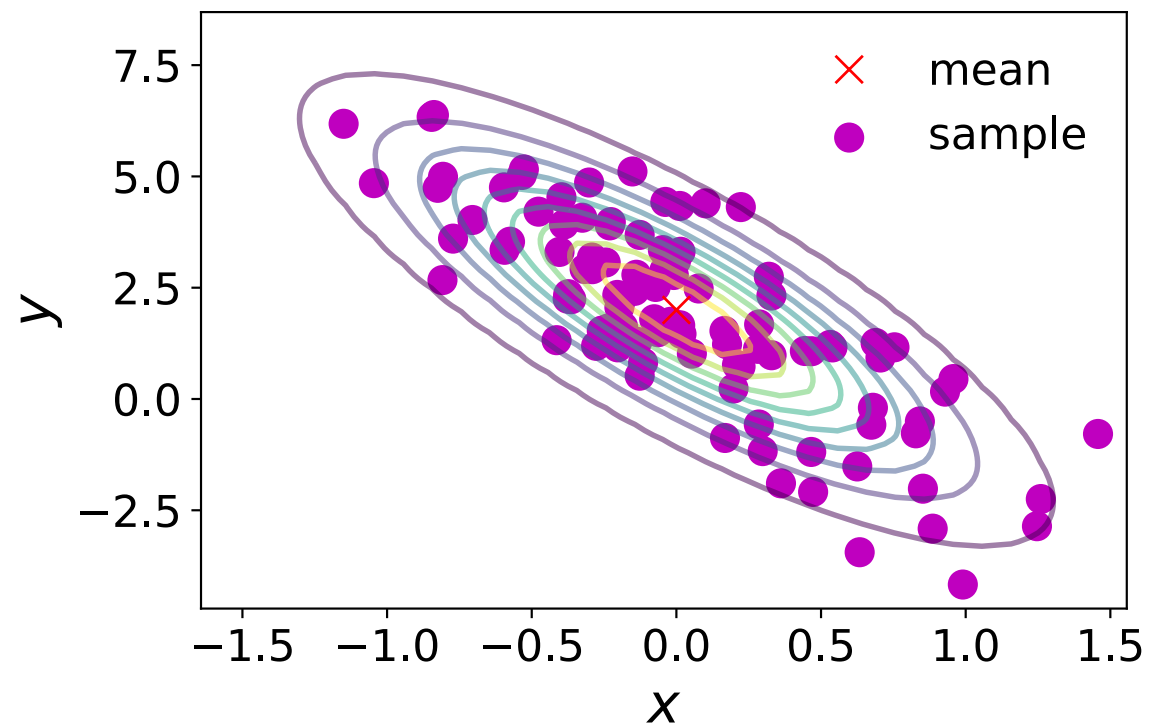


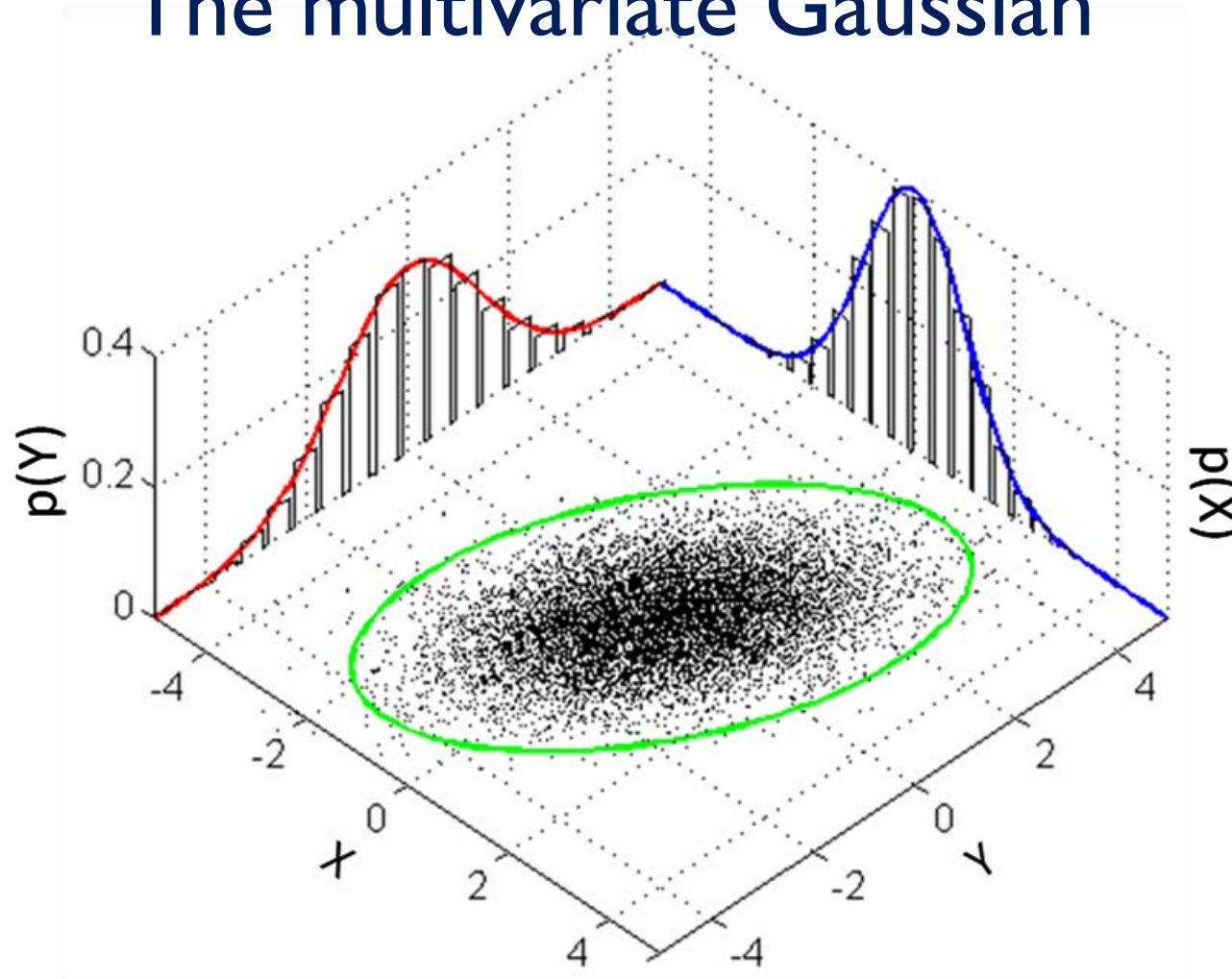
Figure 2.8 Illustration of the effect of outliers on fitting Gaussian, Student and Laplace distributions. (a) No outliers (the Gaussian and Student curves are on top of each other). (b) With outliers. We see that the Gaussian is more affected by outliers than the Student and Laplace distributions. Based on Figure 2.16 of (Bishop 2006a). Figure generated by `robustDemo`.

The multivariate Gaussian



$$p(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-\frac{D}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

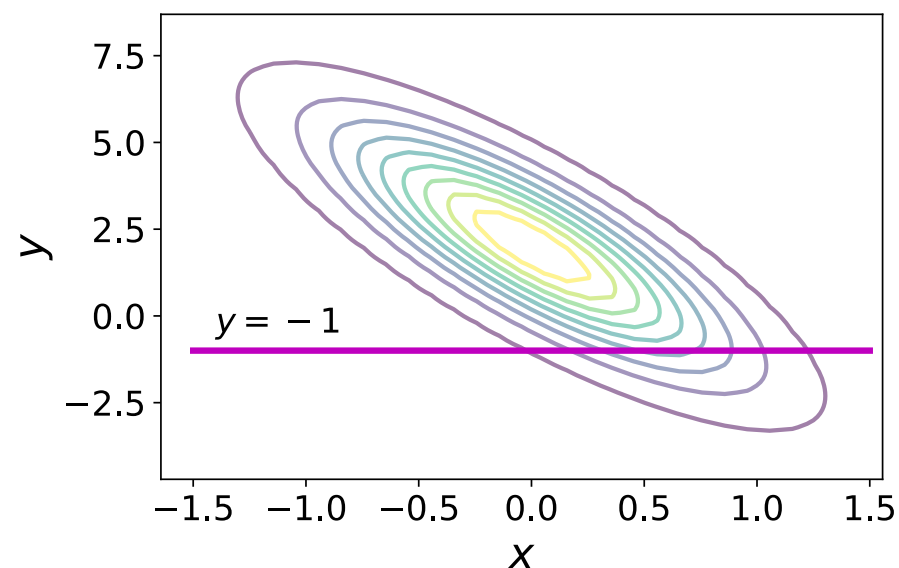
The multivariate Gaussian



Notation	$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
Parameters	$\boldsymbol{\mu} \in \mathbf{R}^k$ — location $\boldsymbol{\Sigma} \in \mathbf{R}^{k \times k}$ — covariance (positive semi-definite matrix)
Support	$\mathbf{x} \in \boldsymbol{\mu} + \text{span}(\boldsymbol{\Sigma}) \subseteq \mathbf{R}^k$
PDF	$\det(2\pi\boldsymbol{\Sigma})^{-\frac{1}{2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$, exists only when $\boldsymbol{\Sigma}$ is positive-definite
Mean	$\boldsymbol{\mu}$
Mode	$\boldsymbol{\mu}$
Variance	$\boldsymbol{\Sigma}$

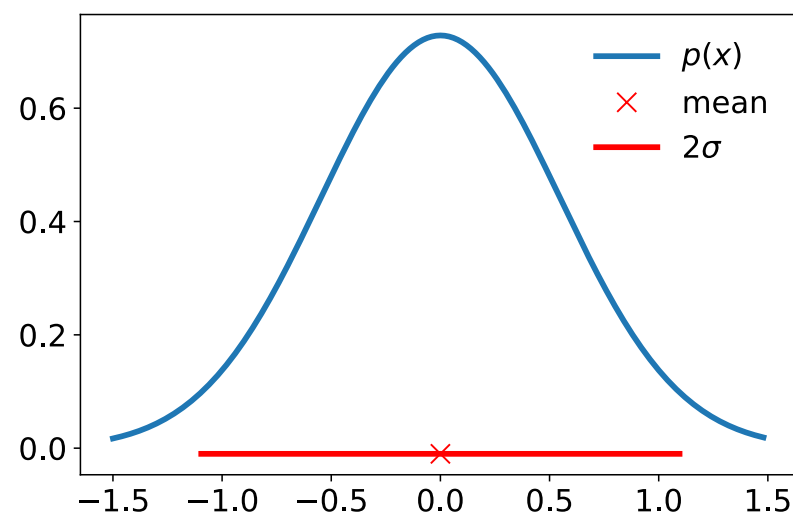
Marginals and conditionals of a Gaussian

$$p(\mathbf{x}, \mathbf{y}) = \mathcal{N} \left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix} \right)$$



Marginal distribution

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{y}) d\mathbf{y} = \mathcal{N}(\mathbf{x} \mid \mu_x, \Sigma_{xx})$$

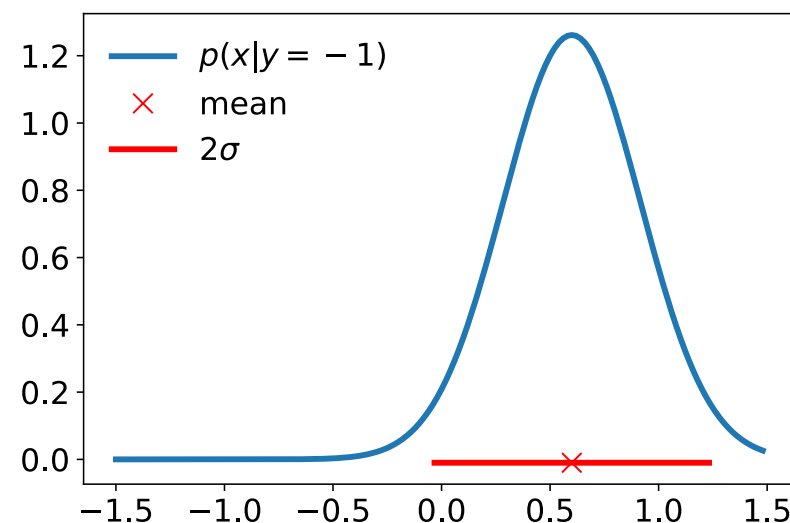


Conditional distribution

$$p(\mathbf{x} \mid \mathbf{y}) = \mathcal{N}(\mu_{x \mid y}, \Sigma_{x \mid y})$$

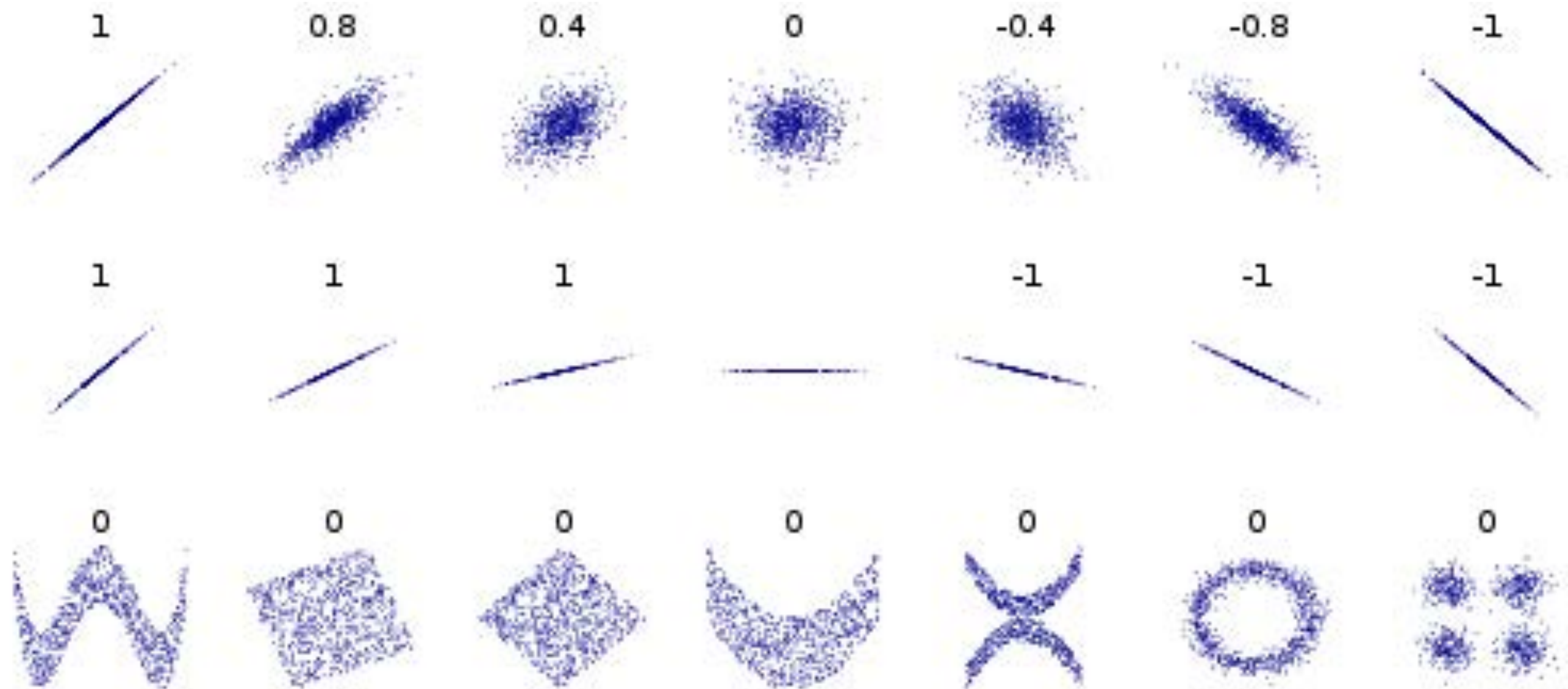
$$\mu_{x \mid y} = \mu_x + \Sigma_{xy} \Sigma_{yy}^{-1} (\mathbf{y} - \mu_y)$$

$$\Sigma_{x \mid y} = \Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}$$



These are unique properties that make the Gaussian distribution very simple and attractive to compute with! It is essentially our main building block for computing under uncertainty.

Correlation and linear dependence



Entropy and Mutual Information

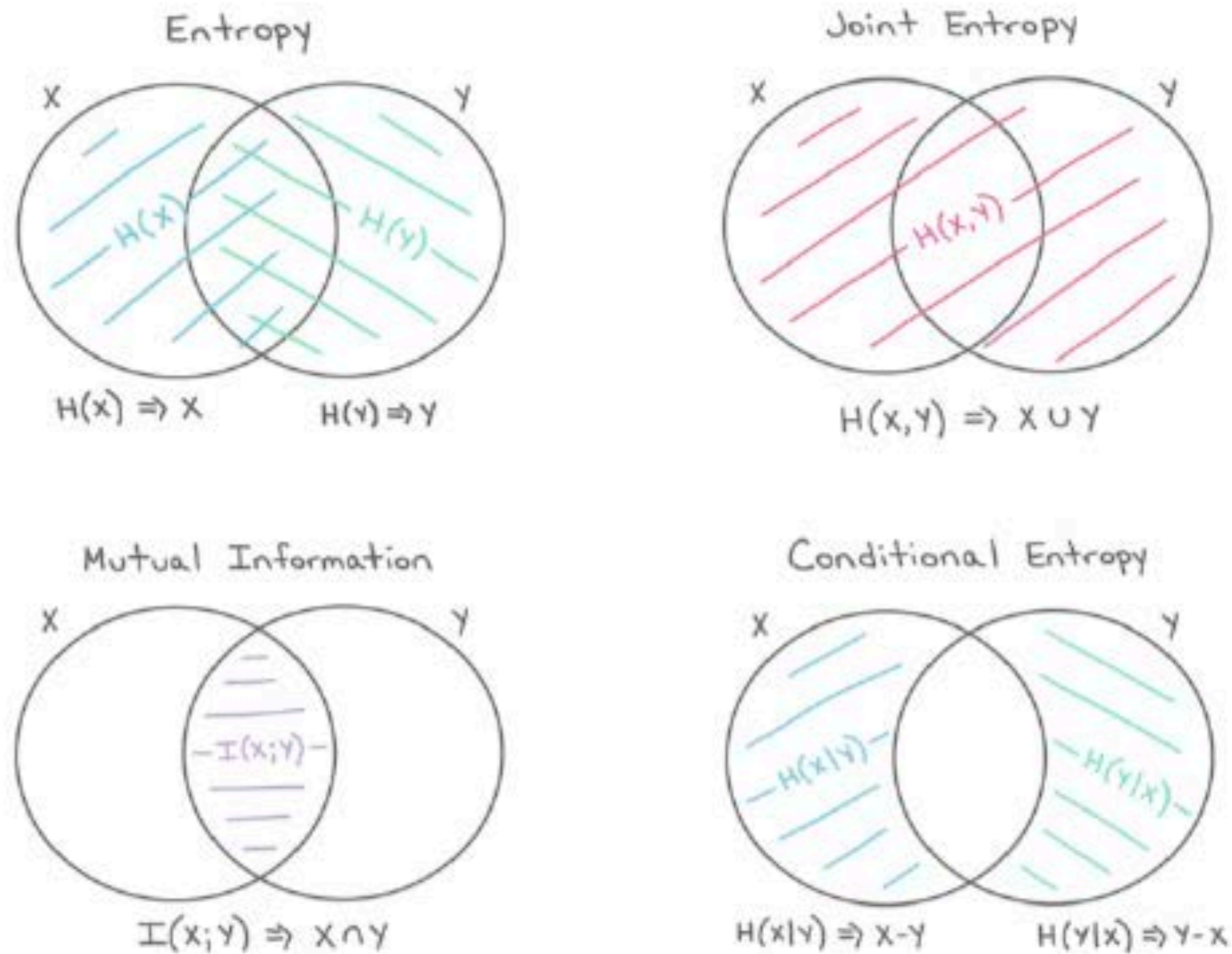


Figure 6.4: The marginal entropy, joint entropy, conditional entropy and mutual information represented as information diagrams. Used with kind permission of Katie Everett.

Forward vs Reverse KL

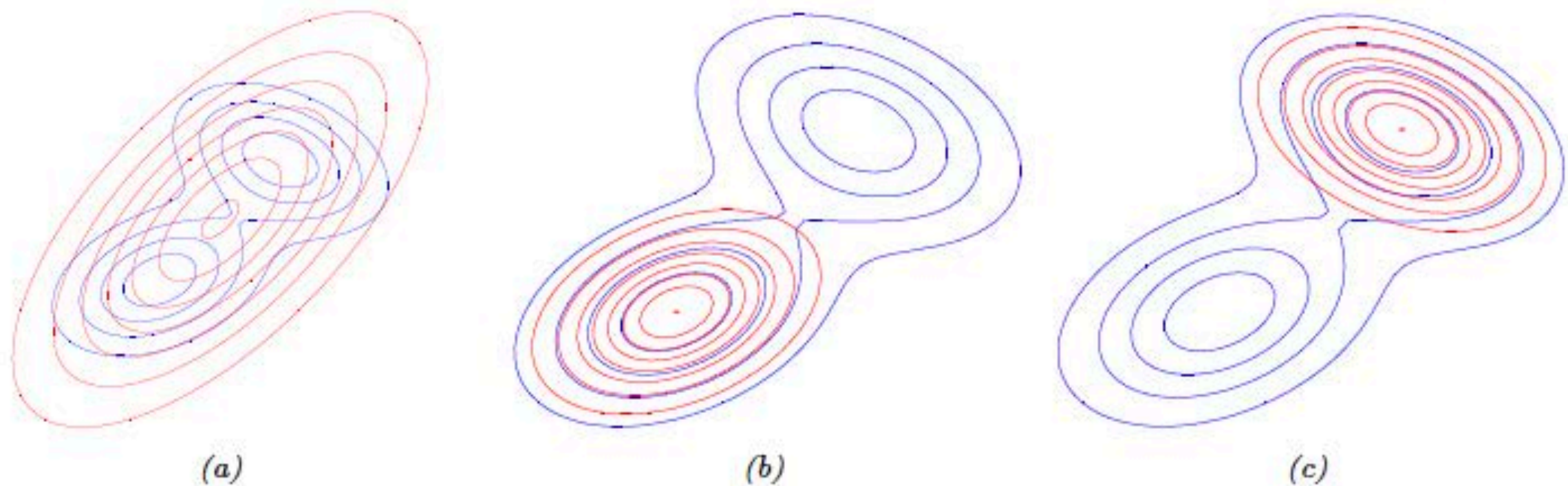
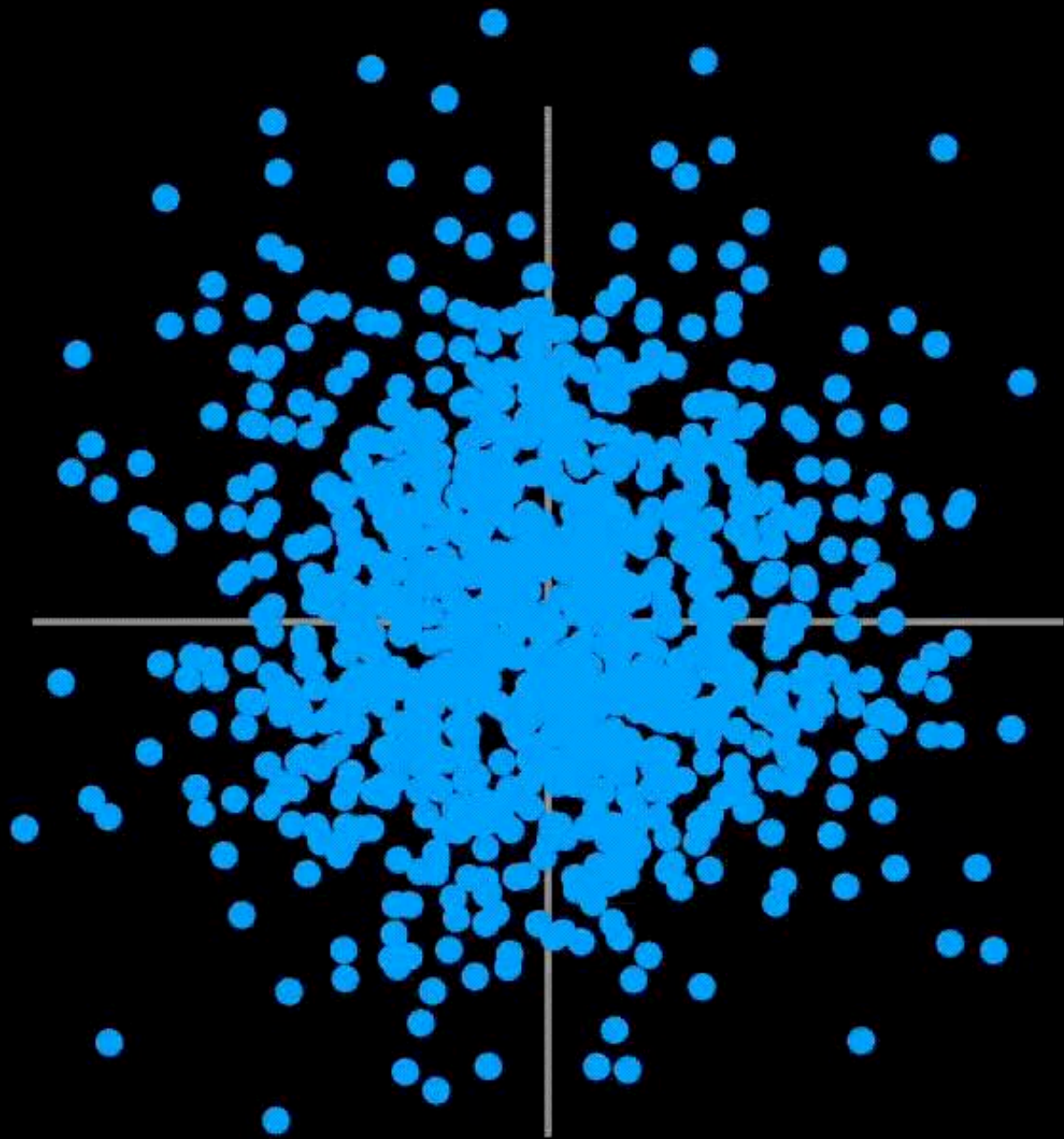


Figure 6.3: Illustrating forwards vs reverse KL on a bimodal distribution. The blue curves are the contours of the true distribution p . The red curves are the contours of the unimodal approximation q . (a) Minimizing forwards KL, $D_{\text{KL}}(p \parallel q)$, wrt q causes q to “cover” p . (b-c) Minimizing reverse KL, $D_{\text{KL}}(q \parallel p)$ wrt q causes q to “lock onto” one of the two modes of p . Adapted from Figure 10.3 of [Bis06]. Generated by `KLfwdReverseMixGauss.ipynb`.

Covariance vs Mutual Information

$$\text{cov}(X, Y) \quad I(X; Y)$$



@ari_seff